

- *amsmath*: Mathematical formula formatting (`\usepackage{amsmath}`)
 - *graphics*: Handling imported images (`\usepackage{graphics}`)
 - *fancyhdr*: Used to insert custom headers (`\usepackage{fancyhdr}`)
 - *makeidx*: Used to build an index for a document (`\usepackage{makeidx}`)
 - *fontenc*: Handling font encodings (`\usepackage[encoding]{fontenc}`)
- Classes and packages are loaded with the following statements:

```
\documentclass [options]{<class>}
\usepackage [options]{<packages>}
```

To get a working LaTeX set-up, you need a LaTeX distribution (TeXLive in GNU/Linux, MikTeX in Windows) and an IDE for creating .tex documents. If you already have LaTeX up and running on your system, then skip the next two sections.

How to get a LaTeX distribution

Most major Linux distributions (Debian, Ubuntu or Fedora) have LaTeX in their repositories—the Texlive distribution is the default for most. On Debian-based systems, you can install it with an *apt-get install texlive*. Or, if you have bandwidth to spare, you could install the whole thing with *apt-get install texlive-full*. However, distro repositories usually have older versions—Debian 5.0 Lenny has TexLive 2007; Squeeze has TexLive 2009. If you want the latest version, download it from the Tux Users Group site (<http://www.tug.org/texlive/>).

LaTeX IDE and compilation

After you create a .tex document, you need to compile it. The easiest way to do both is to use an IDE, which has built-in options for spell-checking, code folding, code completion, compilation and previewing. If you don't have a LaTeX IDE installed, you would use the traditional way of compilation—which is to invoke the LaTeX command in a terminal, like *latex test.tex*. This will create a *dvi* file (and some others). You can view the output with *xdvi test.dvi*. To get a PDF output directly, you can run *pdflatex test.tex*. If you have an IDE, though, all these commands are done by it via GUI options.

There are lots of IDEs available for Windows and GNU/Linux; some major ones for Linux are Emacs + AUCTEX, Kile (KDE version), TexMaker, etc. Vim and Gedit have got some nice LaTeX plug-ins too.

An introduction to Beamer

If you installed TexLive minimal, you may not have the *beamer* class installed—so install it with *apt-get install latex-beamer*. Creating presentations with Beamer is simple if you know how to create documents with LaTeX. Here's a simple Beamer file (results shown in Figure 1):



Figure 1: An introduction to Beamer

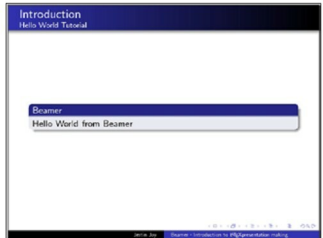


Figure 2: Block

```
\documentclass{beamer}
\usetheme{Warsaw}
\title{Beamer - Introduction to \LaTeX presentation making}
\author{Jestin Joy}
\begin{document}
\begin{frame}
\titlepage
\end{frame}
\begin{frame}
\frametitle{Introduction}
\framesubtitle{Hello World Tutorial}
Hello World.
\end{frame}
\end{document}
```

The first line specifies the document class to use (*beamer*). Packages can also be used in a beamer class. The theme is specified by *\usetheme{Warsaw}*. Like a LaTeX document, the Beamer presentation is enclosed within `\begin{document} - \end{document}` environment. A `\begin{frame}` is followed by the contents of the frame, and then by `\end{frame}`. The first frame above specifies the title slide, which contains the title and author name, with the given theme style. The next frame is the second slide, whose contents are obvious from the code.

Beamer's power is not limited to nicely formatted slides—you can have transitions and animation too. Also, almost all LaTeX constructs can be used inside the beamer class. The commonly used ones are `\begin{itemize}` and `\begin{enumerate}` environments, which provide bullets and